Secrecy Analysis and Optimization of UAV-Assisted IoT Networks With RF-EH and Imperfect Hardware

Gaurav Kumar Pandey, *Student Member, IEEE*, Devendra Singh Gurjar, *Senior Member, IEEE*, Suneel Yadav, *Senior Member, IEEE*, Dragana Krstic, and Yuming Jiang, *Senior Member, IEEE*

***Abstract*—This article introduces uncrewed aerial vehicles (UAVs) assisted communication system for energy-constrained Internet of Things (IoT) devices. Herein, a power beacon transmits radio frequency (RF) signals to enable energy harvesting at the IoT devices (IoDs). The source IoD utilizes harvested energy to transmit information to the destination IoD through multiple UAV relays. Among multiple UAV relays, a selected UAV applies a decode-and-forward operation to relay the information to the destination. An eavesdropper near the destination IoD poses a threat to information secrecy. Consequently, the selected UAV injects artificial noise to degrade the accessibility of the eavesdropper's link. For this setup, we employ a composite fading channel model by combining log-normal distribution for shadowing effects and Nakagami-*m* distribution for small-scale fading to evaluate system performance. Specifically, the compact expressions for secrecy outage probability and effective secrecy throughput (EST) are derived using the Gauss-Hermite quadrature method. Moreover, we conduct a tradeoff analysis between reliability and security by factoring in outage and intercept probabilities. Lastly, we formulate an EST maximization problem by optimizing the system parameters and UAV's locations under the outage and secrecy constraints. We propose a particle swarm optimization and continuous genetic algorithm-based hybrid algorithm for solving the nonlinear optimization problem. The numerical findings demonstrate that the proposed algorithm outperforms the baseline algorithms regarding the achieved EST.**

***Index Terms*—Artificial noise (AN), energy harvesting (EH), Internet of Things (IoT) devices, optimization technique, relay selection, security-reliability tradeoff (SRT), uncrewed aerial vehicles (UAVs).**

I. Introduction

IN RECENT years, advanced communication technologies and emerging concepts, such as artificial intelligence, Internet of Things (IoT), and virtual reality have been revolutionized, making them suitable for various emerging practical applications [1]. According to the Ericsson Mobility Report, IoT connections are projected to increase from 15.7 billion in 2023 to 38.9 billion by 2029 with a compound annual growth rate of 16% [2]. IoT networks can bridge the physical and virtual worlds by integrating numerous intelligent, small, and widely deployable sensing devices [3]. Conversely, some IoT devices (IoDs) can also be deployed in hard-to-reach, remote, and challenging areas, such as oil-rig industries, borders, and natural or artificial disaster-affected regions. On another front, uncrewed aerial vehicle (UAV)-assisted communication has shown immense potential to enhance connectivity in various scenarios, e.g., remote or congested areas, disaster relief operations, and emergencies where establishing conventional infrastructure is challenging [4]. Several attributes of the UAV-enabled networks, such as rapid deployment, scalability, and flexibility in adapting to dynamic communication demands, make them suitable preferences for diverse outdoor environments. In particular, when acting as aerial nodes, UAVs can collect data from the IoDs and relay it efficiently to the destination nodes [5]. Additionally, deploying multiple UAVs can enhance network scalability and performance in various application scenarios.

Realizing communication for these low-power IoDs with the help of UAVs poses several challenges. For example, when the UAV is deployed to enable communication for these IoDs, unauthorized nodes may exploit the availability of line-of-sight (LoS) links to access sensitive information transmitted through air-to-ground (A2G) links. Moreover, the broadcast nature of the wireless channels can pose severe security threats to UAV communications [6]. Traditional cryptographic algorithms are commonly utilized to enhance the security of information transfer (IT) in communication networks. However, advances in the computational capabilities of terrestrial eavesdroppers can introduce inherent vulnerabilities in cryptographic systems, particularly in the distribution and management of secret keys. To address these challenges, physical layer (PHY) security has emerged as an effective solution to improve secure information transmission over wireless channels [7].

From the PHY security perspective, attacks on UAV-integrated wireless networks can be categorized into two main scenarios; first, when the terrestrial malicious nodes can impose eavesdropping or jamming attacks on the UAV's A2G link; second, when the malicious UAV launches eavesdropping or jamming attacks on legitimate nodes. Various

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Gaurav Kumar Pandey and Devendra Singh Gurjar are with the Department of Electronics and Communication Engineering, National Institute of Technology Silchar, Silchar 788010, India (e-mail: gaurav_rs@ece.nits.ac.in; dsgurjar@ece.nits.ac.in).

Suneel Yadav is with the Department of Electronics and Communication Engineering, Indian Institute of Information Technology Allahabad, Prayagraj 211015, India (e-mail: suneel@iiita.ac.in).

Dragana Krstic is with the Faculty of Electronic Engineering, University of Nis, Niš 18104, Serbia (e-mail: dragana.krstic@elfak.ni.ac.rs).

Yuming Jiang is with the Department of Information Security and Communication Technology, Norwegian University of Science and Technology, 7491 Trondheim, Norway (e-mail: yuming.jiang@ntnu.no).

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techniques like relay selection, cooperative jamming, UAV's path planning, and injecting artificial noise (AN) are popularly utilized in the literature to improve data secrecy in UAV communications [8]. Moreover, combining PHY security with traditional cryptographic methods can safeguard sensitive information transmission against increasingly sophisticated security threats.

In addition to the challenges of inaccessibility and security, limited operational lifespan is another crucial concern for IoDs. The lifetime of these IoDs is curtailed as small-sized batteries with limited energy resources usually power them. This hampers the data gathering and dissemination capabilities of the IoDs as they require frequent charging or battery replacement. The IoDs deployed in nonaccessible areas may need alternative power sources where the conventional methods become inefficient and infeasible [9]. Numerous energy harvesting (EH) methods have emerged to tackle the energy limitation problem and improve the functionality of IoDs [10]. Much emphasis has been put on simultaneous wireless information and power transmission (SWIPT) to enable self-sustaining wireless networks in the last few years [11]. Ambient and dedicated radio frequency (RF) sources can enable RF-EH and make UAV-assisted IoT networks self-sustaining, thus improving the network's durability [6]. However, the challenges in RF-EH, such as low power density, significant temporal variations in power levels, and the complexity of designing impedance matching for dynamic systems, must be handled before its implementation. In this direction, several researchers in [12], [13], and [14] have implemented efficient rectenna designs to counter the above-mentioned challenges in RF-EH at low-power IoD nodes. Another challenge arises due to the hardware imperfections associated with RF transceivers due to nonlinearities, phase noise, and imperfections in analog-to-digital converters in the communicating nodes, which can severely hamper the performance of UAV-assisted networks. Hence, it is essential to account for hardware impairments (HIs) when assessing the system performance accurately [15].

A. Prior Arts

Several works [16], [17], [18], [19] have leveraged PHY security techniques to secure data transfer in UAV communications. Specifically, the Wang et al. [16] and Li et al. [17] maximized the secrecy performance using the UAV-assisted AN injection method. Whereas Zhang et al. [18] improved the covertness of full-duplex (FD) UAV relay-assisted networks through flexible UAV deployment. Similarly, Xu et al. [19] employed relay selection to counter eavesdropping in cognitive decode-and-forward (DF) relay networks. Nevertheless, Wang et al. [16], Li et al. [17], Zhang et al. [18], and Xu et al. [19] did not consider the EH techniques in their system models. Further, Salem et al. [20] employed amplify-and-forward (AF)-based relay selection, whereas Wang et al. [21] employed optimal antenna selection to improve data secrecy. The works [16], [17], [18], [19], [20], [21] primarily focused on the security of UAV-assisted networks without addressing the reliability of legitimate communication. To analyze the security-reliability tradeoff (SRT), Cao et al. [22] utilized AN and FD relay selection. On the other hand, Hoang et al. [23] formulated the SRT as a stochastic-constrained optimization problem and solved it using deep neural networks.

Unlike [22] and [23], Ding et al. [24] analyzed SRT by incorporating jammer-aided relay selection with imperfect channel estimations. Similarly, Nguyen et al. [25] introduced a friendly jammer to enhance security against eavesdroppers and emphasized SRT as a critical metric. However, these studies [22], [23], [24], [25] did not provide a comprehensive view of security and reliability to offer insights into achievable system throughput. In contrast, Sun et al. [26] proposed an adaptive genetic simulated annealing algorithm to improve effective secrecy throughput (EST) for UAV-assisted communication. Likewise, Cao et al. [27] investigated EST performance in uplink nonorthogonal multiple access systems, whereas Yan et al. [28] conducted SRT analysis in cooperative cognitive radio systems, comparing energy-aware and nonenergy-aware relay selection schemes. Nevertheless, these works [27], [28] utilized terrestrial relays in their considered system. Apart from these studies, significant attention has been given to the practical challenges in UAV communications by jointly analyzing security and reliability issues [29], [30], [31], [32]. Specifically, Kong et al. [29] compared interference-aware and noninterference relay selection schemes for SRT analysis in the presence of HIs over Nakagami-m fading channels.

Nguyen et al. [30] jointly minimized the outage probability (OP) and intercept probability (IP). In a similar vein, Diao et al. [31] proposed dynamic power allocation and UAV-based jamming schemes. Additionally, Yuan et al. [32] explored SRT in UAV-aided networks that utilize reconfigurable intelligent surfaces. However, Sun et al. [26], Kong et al. [29], Diao et al. [31], and Yuan et al. [32] did not incorporate the RF-EH technique. Numerous optimization techniques have been used in the literature to obtain optimal UAV positions and resource allocations that maximize network secrecy [33]. Specifically, algorithms, such as simulated annealing, genetic algorithms (GA), and particle swarm optimization (PSO) are widely employed [34]. Yan et al. [35] combined PSO and GA by replacing traditional crossover and mutation phases with two movements, one toward the global best candidate and the other toward the candidate's historical best. Similarly, Tariq et al. [36] proposed a genetic local particle-based hybrid algorithm, combining PSO properties with additional features from GA. The optimization problems in [33], [34], [35], and [36] focused on maximizing secrecy or reliability performance metrics. Note that, the combined impact of HIs and composite channel fading on the secrecy and reliability of multi-UAV-assisted networks remains unexplored under challenging scenarios.

B. Motivation and Contributions

Motivated by the existing literature and identified gaps, we focus on establishing a multiple UAVs-aided relay network for the energy-constrained IoD pairs in severely obstructed or shadowed environments where direct communication links are not present. Additionally, terrestrial eavesdroppers in the

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TABLE I
COMPARISON OF PROPOSED WORK WITH RELEVANT LITERATURE WORKS

Works	System Model				AN Injection or Jamming	EST	Optimizing	
	Multi UAV	Hardware Impairments	RF-EH	Power Beacon			System Parameters	UAV Attributes
[18]	×	×	×	×	Jamming	×	✓	×
[19]	✓	×	×	×	Jamming	×	✓	×
[20]	✓	×	✓	×	×	×	×	×
[22]	✓	×	×	×	AN injection	×	×	×
[25]	×	×	✓	×	Jamming	✓	×	✓
[29]	✓	✓	×	×	×	✓	×	✓
[30]	×	×	✓	✓	AN injection	×	✓	×
[31]	✓	×	×	×	Jamming	✓	×	×
[33]	✓	×	×	×	×	×	✓	×
[34]	✓	×	✓	✓	×	×	✓	✓
This Work	✓	✓	✓	✓	AN injection	✓	✓	✓

network can get access to the IT over the UAV's A2G link. Therefore, we utilize the UAV-based relay selection technique for relaying to improve reliability and inject AN to degrade the wiretapping link of the eavesdropper. The heavily shadowed A2G and G2A links are modeled with a composite channel fading model, incorporating log-normal and Nakagami-m distributions for shadowing and small-scale fading, respectively. We provide Table I to highlight the contributions made in this work and compare them with relevant studies. A brief overview of this article's significant contributions is as follows.

1) We derive the signal-to-noise-plus-distortion ratio (SNDR) of the A2G and G2A links based on the received signals at the corresponding nodes in the presence of HIs. Then, we perform an SNDR-based relay selection to find the best UAV among multiple UAVs to provide relay assistance.
2) We utilize UAV-based AN injection to enhance the network's security, wherein the UAV injects AN with the decoded signal based on the power allocation factor. For this setup, we derive the accurate expression of secrecy OP (SOP) and perform SRT analysis by obtaining the expressions of OP and IP using the Gauss-Hermite quadrature method.
3) Moreover, we utilize OP expression and Wyner's coding theory-based SOP formulation to obtain the exact EST expressions and address the EST maximization problem. We optimize the UAV's position, EH, and system parameters while considering the outage and secrecy constraints.
4) Lastly, we employ a continuous GA (CGA)-PSO-based hybrid algorithm for solving the nonlinear optimization problem and compare the results with baseline algorithms, specifically individual PSO and GA, as well as hybrid algorithms as in [35] and [36]. We also demonstrate the effectiveness of combining relay selection with AN injection methods compared to the conventional technique in the numerical results section.

Notations: The probability, cumulative distribution function (CDF), and probability density function (PDF) of a random variable X are denoted as $\Pr[\cdot]$, $F_X(\cdot)$, and $f_X(\cdot)$, respectively. The complete, upper, and lower incomplete gamma functions are given by $\Gamma(\cdot)$, $\Gamma(\cdot,\cdot)$ and $\Upsilon(\cdot,\cdot)$. Modified Bessel function of the second kind of ν_{th} order is given as $\mathcal{K}_\nu(\cdot)$, and Kummer hypergeometric function is denoted by $\Psi(\cdot,\cdot;\cdot)$.

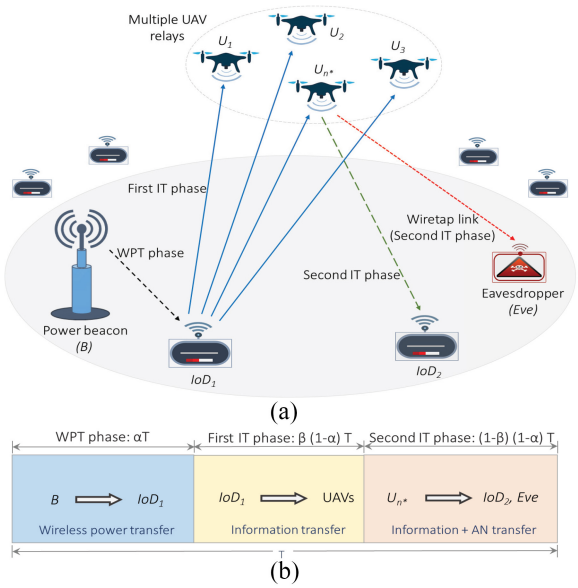

Fig. 1. (a) Multiple UAV-assisted secure IoD communication. (b) Proposed transmission block for EH and IT.

II. SYSTEM MODEL DISCUSSION

A. System Model

As shown in Fig. 1(a), we consider a wireless network consisting of energy-constrained IoDs communicating through multiple UAV relays to ensure reliable and secure IT in the presence of a terrestrial eavesdropper [37]. Energy-constrained IoDs use a nearby power beacon to harvest energy to transmit data, while the UAVs rely on their onboard batteries. These batteries provide the necessary power to UAVs for performing both relaying and broadcasting AN [15], [18], [22], [25], [30], [31]. Although multiple IoDs can be deployed in a small area, we emphasize facilitating IT between a specific pair of IoDs. The involved nodes, such as power beacon (B), nth UAV among N number of UAVs (U_n) $\forall n \in N$, an IoD pair (IoD_1 and IoD_2), and an eavesdropper (Eve) are donated with subscripts B, 1, 2, U_n, and E. The horizontal distances between the communicating nodes are given as $r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$ $\forall$ $(i, j) \in \{(B, 1), (1, U_n), (U_n, 2), (U_n, E)\}$. Moreover, the distances and

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elevation angles between U_n and the terrestrial nodes are computed as $d_{U_n,\iota} = \sqrt{r_{U_n,\iota}^2 + H_{U_n}^2}$ and $\theta_{U_n,\iota} = \arctan(H_{U_n}/r_{U_n,\iota})$, respectively $\forall \iota \in \{1, 2, E\}$, where H_{U_n} is the flying altitude of the U_n.

In this context, the primary objective is to facilitate secured IT between IoD_1 and IoD_2 via multiple UAV relays. As depicted in Fig. 1(b), the overall transfer of energy and information is executed in the data transmission block of duration T, segmented into three phases representing the WPT and two IT phases. Specifically, in WPT phase of αT duration, IoD_1 harnesses energy from the RF signal transmitted by a nearby B. Here, α represents the TS factor, such that $0 < \alpha < 1$. For analytical treatment in conducting the SRT analysis, we adopt a linear EH model for the energy-constrained IoD_1 [16], [18], [19], [21], [27], [28], [30]. Subsequently, in the first IT phase, denoted as $\beta(1-\alpha)T$, where β represents the IT time factor such that $0 < \beta < 1$, IoD_1 transmits information to multiple UAVs. Then, based on the instantaneous SNDR obtained in the first IT phase, a UAV relay is selected to forward the signal to IoD_2 using DF relaying method.

A terrestrial Eve is present near IoD_2 and can potentially wiretap information through the UAV's A2G channel. Therefore, in the second IT phase, expressed as $(1-\beta)(1-\alpha)T$, the selected UAV introduces self-generated AN into the decoded data to enhance its security before relaying it to IoD_2. We assume that the channel state information (CSI) of the legitimate links is known to the UAVs [29]. Moreover, the legitimate nodes can obtain the CSI of Eve[1] using some advanced feasible techniques [30], [38]. Such assumptions are widely utilized in [29], [30], [38], [39], [40] while assessing the security performance of their considered systems.

B. Channel Model

Integrating UAVs into communication networks results in diverse channel conditions mainly affected by shadowing and small-scale fading. We employ a composite channel model that combines the Nakagami-m and log-normal distributions to capture these effects under a block fading scenario. This model effectively characterizes the small-scale fading and shadowing in A2G and G2A communication links between multi-UAVs and terrestrial nodes [41], [42], [43]. In this work, we derive accurate PDF expressions for composite fading distributions in scenarios involving dual communication links. Composite fading channel's PDF, denoted as $f_X(x)$, can be derived by integrating the PDF of log-normal distribution, denoted as $f_W(w)$, for characterizing the shadowing, and gamma distribution' PDF for representing the Nakagami-m faded signal amplitude [41]. Therefore, the PDF of the composite fading channel is expressed as $f_X(x) = \int_0^\infty f_X(x/w) f_W(w) dx$, and obtained as

$$f_X(x) = \int_0^\infty \frac{1}{\Gamma[m_{i,j}]} \left(\frac{m_{i,j}}{w}\right)^{m_{i,j}} \frac{10\, x^{m_{i,j}-1} \mathrm{e}^{-\left(\frac{m_{i,j}x}{w}\right)}}{\log_e 10 \sqrt{2\pi}\delta_{i,j} x} \times \mathrm{e}^{-\left(\frac{10\log_{10} x - \mu_{i,j}}{2\delta_{i,j}^2}\right)^2} dx \tag{1}$$

where, $\delta_{i,j}$ is the standard deviation for shadowing and given as $\delta_{i,j} = c_1 \mathrm{e}^{-\theta_{i,j} d_1}$ wherein the environment parameters c_1 and d_1 varies for different scenarios, and $m_{i,j}$ is the fading severity parameter. After performing some mathematical operations and the Gauss-Hermite quadrature numerical technique [42], (1) can be simplified and the final PDF of the composite fading channel is given as

$$f_X(x) = \frac{1}{\sqrt{\pi}} \sum_{a=1}^{A} w_A \left(\frac{m_{i,j}}{\mathcal{A}_{i,j}}\right)^{m_{i,j}} \frac{x^{m_{i,j}-1}}{\Gamma[m_{i,j}]} e^{-\left(\frac{m_{i,j}x}{\mathcal{A}_{i,j}}\right)} \tag{2}$$

where $\mathcal{A}_{i,j} = 10^{(\sqrt{2}\delta_{i,j} t_a + \vartheta_{i,j})/10}$, w_a denotes the Gauss-Hermite quadrature weights and the Hermite polynomial roots are given by $[t_a]_{a=1}^{A}$ with A samples. Moreover, we incorporate average path loss for the given links, which can be derived using LoS and non line-of-sight (NLoS) probabilities. It is denoted as $\vartheta_{i,j}$ and depends on elevation angle and environmental factors [42]. Mathematically, $\vartheta_{i,j} = (\mathcal{P}_{\text{LoS}}(\theta) \times \vartheta_{\text{LoS}} + \mathcal{P}_{\text{NLoS}}(\theta) \times \vartheta_{\text{NLoS}})$, where $\mathcal{P}_{\text{LoS}}(\theta) = 1/(1 + a\mathrm{e}^{-b(\theta_{i,j}-a)})$ and $\mathcal{P}_{\text{NLoS}}(\theta) = (1 - \mathcal{P}_{\text{LoS}}(\theta))$ represent the LoS and NLoS probabilities, where a and b are representing the environment-based variables. Furthermore, ϑ_{LoS} and ϑ_{NLoS} denotes the LoS and NLoS components' mean path-loss that are obtained as $\vartheta_{\text{LoS}} = 20\log_{10}(4\pi f_c/c) + 20\log_{10} d_{i,j} + \psi_{\text{LoS}}$ and $\vartheta_{\text{NLoS}} = 20\log_{10}(4\pi f_c/c) + 20\log_{10} d_{i,j} + \psi_{\text{NLoS}}$, where f_c and c denotes the carrier frequency and light's speed, ψ_{LoS} and ψ_{NLoS} are propagation environment constants [42].

C. SNDR Expressions

Accounting for the impact of HIs, the received signal at node j from node i is expressed as $y_{i,j} = h_{i,j}(\sqrt{P_i}x_i + \zeta_{i,j}) + n_j$, where $\zeta_{i,j} \sim \mathcal{CN}(0, k^2 P_i)$ represents the distortion noise caused by transceiver impairments of communicating nodes i and j. Here, P_i signifies the transmit power, and $k = \sqrt{k_i^2 + k_j^2}$ is a composite parameter with k_i and k_j denoting HIs at the transmitter and receiver, respectively. The scenario with $\zeta_{i,j} = 0$ corresponds to an ideal case, i.e., without any HIs. During the WPT phase, IoD_1 harvests the energy from the B's transmitted RF signal with transmit power given by P_B. Then, harvested energy by IoD_1 is given as IoD_1, where η is the energy conversion efficiency (0.7–0.95), and $h_{B,1}$ represents the channel coefficient [14]. Therefore, the power available at IoD_1 is expressed as

$$P_1 = \frac{\eta\alpha\left(P_B |h_{B,1}|^2\right)}{\beta(1-\alpha)}. \tag{3}$$

On utilizing the harvested power, IoD_1 transmits the signal to multiple UAV relays in the first IT phase. Thus, the signal received at U_n in the presence of HIs is expressed as $y_{1,U_n} =$

[1]Here, we consider the worst case where the eavesdropper has high computational capabilities and can try to estimate CSI perfectly by analyzing intercepted signals, such as pilot signals or reference symbols, transmitted during channel estimation. This can be achieved by applying advanced techniques, such as machine learning and adaptive filtering, especially when the eavesdropper has access to sophisticated computational resources. However, there may be conditions/scenarios where the eavesdropper cannot estimate the CSI accurately, which requires further consideration.

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$(\sqrt{P_1}z_1 + \zeta_{1,U_n})h_{1,U_n} + n_{U_n}$, where z_1 and P_1 are unit energy symbol and transmit power of the IoD_1's signal, and $n_{U_n} \sim \mathcal{CN}(0, N_0)$ denotes the additive white Gaussian noise (AWGN) at U_n. Therefore, the SNDR at the U_n based on the received signal is given as

$$\gamma_{1,U_n} = \frac{\eta\alpha P_B|h_{B,1}|^2|h_{1,U_n}|^2}{\eta\alpha k^2 P_B|h_{B,1}|^2|h_{1,U_n}|^2 + \beta(1-\alpha)N_0}. \tag{4}$$

Based on the SNDR expression given in (4), an optimal IoD_1-U_n relay link is selected by incorporating the SNDR-based relay selection technique [19], [20], [24], [28], [29]. This can be mathematically expressed as

$$n^* = \arg \max_{n\in\{1,\ldots,N\}} \{\gamma_{1,U_n}\}. \tag{5}$$

The selected UAV relay, denoted as U_{n^*}, is considered to have the best channel conditions, and therefore, it is assumed to decode the information signal successfully which is received from IoD_1 in the first IT phase [44], [45]. Next, before broadcasting the information signal, the UAV introduces a self-generated AN to prevent the Eve from accessing the confidential information. The power allocation factor, denoted as $0 < \epsilon \leq 1$, determines the proportion of power allocated to the information signal and the AN. Selecting an optimal ϵ value ensures efficient utilization of the U_{n^*}'s available power, balancing the energy dedicated to the information signal and AN injection. Therefore, the signals received at the legitimate node IoD_2 and *Eve* are given as

$$y_{U_n^*,l} = \Big(\underbrace{\sqrt{\epsilon P_{U_{n^*}}}z_{U_{n^*}}}_{\text{Information signal}} + \underbrace{\sqrt{(1-\epsilon)P_{U_{n^*}}}z_{\text{A}}}_{\text{AN}} + \zeta_{U_{n^*},l}\Big)h_{U_{n^*},l} + n_l \tag{6}$$

where $l \in \{2, E\}$, $P_{U_{n^*}}$ and $z_{U_{n^*}}$ represent the transmit power and unit energy symbol of U_{n^*}. Herein, $x_{\text{A}} \in \mathcal{CN}(0, 1)$ denotes the normalized complex Gaussian AN transmitted by the U_{n^*}, whereas the AWGN at IoD_2 and Eve is represented by $n_l \sim \mathcal{CN}(0, N_0)$ [16], [22]. Notably, the AN introduced by U_{n^*} can be eliminated by IoD_2, and therefore, it can retrieve the information from the received signal. However, Eve cannot remove the AN from the received signals [18], [19], [32]. This assumption holds because the AN generated by U_{n^*} is produced using pseudo-random sequences [27]. The stochastic nature of the seed generating these sequences increases the complexity [45], making it practically impossible for Eve to decode the signal even with long-term observation of legitimate transmissions. The instantaneous SNDRs at the IoD_2 and Eve based on the signal received in (6) is given by

$$\gamma_{U_n,2} = \frac{\epsilon P_{U_n}|h_{U_n,2}|^2}{\epsilon k^2 P_n|h_{U_n,2}|^2 + N_0} \tag{7}$$

$$\gamma_{U_n,E} = \frac{\epsilon P_{U_n}|h_{U_n,E}|^2}{(1-\epsilon+k^2)P_{U_n}|h_{U_n,E}|^2 + N_0}. \tag{8}$$

The SNDR expressions obtained in (4), (7), and (7) are utilized to derive the performance metrics, that can offer insightful remarks into the system's behavior.

III. Performance Analysis

In this section, we derive the closed-form expressions of the SOP by using the instantaneous SNDRs. Followed by an analysis of SRT performance in terms of OP and IP. Then, we obtain the expression for EST by using OP and Wyner's coding theory-based SOP.

A. Secrecy Outage Probability

Generally, SOP is considered as a crucial metric for assessing the secrecy performance of the system. It is characterized as an event which occurs when the secrecy capacity $\mathcal{C}$s falls below a predefined target rate, $\mathcal{R}_{\text{th}}$ (bps/Hz) [7]. Specifically, secrecy capacity is mathematically expressed as the difference between the instantaneous capacity of the legitimate link and eavesdropper's wiretapping link, i.e., $\mathcal{C}_{\text{s}} = (\mathcal{C}_{U_n,2} - \mathcal{C}_{U_n,E})$, where $\mathcal{C}_{U_n,2} = (1-\beta)(1-\alpha)\log_2(1+\gamma_{U_n,2})$ and $\mathcal{C}_{U_n,E} = (1-\beta)(1-\alpha)\log_2(1+\gamma_{U_n,E})$. Given that the UAV successfully decodes the information in the first IT phase [19], [21], [22], SOP is mathematically expressed as

$$\begin{aligned}\mathcal{P}_{\text{sop}} &= \Pr[\mathcal{C}_{\text{s}} < \mathcal{R}_{\text{th}}] = \Pr\left[\frac{1+\gamma_{U_n,2}}{1+\gamma_{U_n,E}} < \phi\right] \\ &\sim eq\Pr\left[\frac{\gamma_{U_n,2}}{\gamma_{U_n,E}} < \phi\right]\end{aligned} \tag{9}$$

where $\phi = 2^{\mathcal{R}_{\text{th}}/(1-\beta)(1-\alpha)}$ is the target SNR of the system. The final expression of SOP is provided in Lemma 1.

Lemma 1: The final expression of $\mathcal{P}_{\text{sop}}$ is given by

$$\begin{aligned}\mathcal{P}_{\text{sop}} = &\frac{1}{\sqrt{\pi}}\sum_{\varrho=0}^{\mathcal{K}}\frac{w_\varrho}{\Gamma[m_{U_n,E}]}\Upsilon\left[m_{U_n,E}, \frac{m_{U_n,E}}{\phi A_2 - A_3\mathcal{A}_{U_n,E}}\right] + \frac{1}{\pi}\sum_{\kappa=0}^{\mathcal{J}} \\ &\times \sum_{l=0}^{L}\frac{w_\kappa w_l}{\Gamma[m_{U_n,E}]}\left[\Gamma\left(m_{U_n,E}, \frac{m_{U_n,E}}{\phi A_2 - A_3\mathcal{A}_{U_n,E}}\right) - \left(\frac{m_{U_n,2}}{\mathcal{A}_{U_n,2}}\right)^p\right. \\ &\times \left(\frac{m_{U_n,E}}{\mathcal{A}_{U_n,E}}\right)^{m_{U_n,E}}\sum_{p=0}^{m_{U_n,2}-1}\frac{(-1)^p}{p!}\left(\frac{1}{\phi A_2 - A_3}\right)^{m_{U_n,E}+p} \\ &\times \sum_{r=0}^{m_{U_n,E}+p-1}\binom{m_{U_n,E}+p-1}{r}\left(\frac{m_{U_n,E}}{\mathcal{A}_{U_n,E}}\frac{m_{U_n,2}}{\mathcal{A}_{U_n,E}\phi}\right)^{\frac{m_{U_n,2}-r}{2}} \\ &\times \mathcal{K}_{m_{U_n,2}-r}\left(2\sqrt{\frac{m_{U_n,E}}{\mathcal{A}_{U_n,E}}\frac{m_{U_n,2}}{\mathcal{A}_{U_n,2}}\frac{\phi}{(\phi A_2 - A_3)^2}}\right) \\ &\left.\times e^{-\left(\frac{1}{\phi A_2 - A_3}\left(\frac{m_{n,e}}{\mathcal{A}_{U_n,E}} - \frac{m_{U_n,2}\phi}{\mathcal{A}_{U_n,2}}\right)\right)}\right].\end{aligned} \tag{10}$$

Proof: Please see Appendix A. ■

Remark 1: The analysis of (10) reveals that the SOP is predominantly dominated by the power terms, modified Bessel function of the second kind, and exponential function. Modified Bessel function of the second kind can be expanded as $\mathcal{K}_\nu(x) = \sqrt{\pi}e^{-x}(2x)^\nu\Psi(\nu+0.5, 1+2\nu; 2x)$ [46, eq. (9.328)]. Now, on substituting this into (10), it can be observed that the SOP primarily relies on power terms and the exponential function. Consequently, the SOP's increasing or decreasing behavior can be easily analyzed based on various channel and system parameters, as further corroborated numerically in Section V.

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B. Security-Reliability Tradeoff

In practice, it is observed that although system reliability improves with the increase in the transmit power, this compromises the security [29]. Achieving an optimal balance between security and reliability becomes crucial under diverse conditions for effective practical applications [23], [32]. Therefore, this section focuses on assessing security and reliability performance by conducting a comprehensive SRT analysis based on OP and IP [25], [28], [47], [48].

1) Outage Probability: OP is characterized as the event where the instantaneous data rate drops below a specified threshold rate, $\mathcal{R}_{\text{th}}$. To determine the OP in DF relaying, data transmission through G2A and A2G links in both the IT phases is taken into account. This is mathematically expressed as

$$\begin{aligned}\mathcal{P}_{\text{out}} &= 1-\left(1-\Pr\left[\mathcal{R}_{1,U_n^*}<\mathcal{R}_{\text{th}}\right]\right)\left(1-\Pr\left[\mathcal{R}_{U_n,2}<\mathcal{R}_{\text{th}}\right]\right)\\ &= 1-\left(1-\Pr\left[\gamma_{1,U_n^*}<\psi\right]\right)\left(1-\Pr\left[\gamma_{U_n,2}<\delta\right]\right)\end{aligned} \tag{11}$$

where $\mathcal{P}_{1,U_n^*}=\Pr[\mathcal{R}_{1,U_n^*}<\mathcal{R}_{\text{th}}]$ and $\mathcal{P}_{U_n,2}=\Pr[\mathcal{R}_{U_n,2}<\mathcal{R}_{\text{th}}]$ represents the successful decoding of the signal by U_{n^*} and IoD_2 in the first and second IT phases, respectively. Herein, $\mathcal{R}_{1,U_n^*}=\beta(1-\alpha)\log_2(1+\gamma_{1,U_n^*})$ and $\mathcal{R}_{U_n,2}=(1-\beta)(1-\alpha)\log_2(1+\gamma_{U_n,2})$, where $\psi=2^{\mathcal{R}_{\text{th}}/\beta(1-\alpha)}-1$ and $\delta=2^{\mathcal{R}_{\text{th}}/(1-\beta)(1-\alpha)}-1$ are the target SNRs of G2A and A2G links in both the IT phases, respectively. On substituting (5) in (11), the OP expression OP is given by

$$\begin{aligned}\mathcal{P}_{\text{out}} &= 1-\left(1-\prod_{n=1}^{N}\Pr\left[\gamma_{1,U_n}<\psi\right]\right)\left(1-\Pr\left[\gamma_{U_n,2}<\delta\right]\right)\\ &= 1-\left(1-(F_{\gamma_{1,U_n}}(\psi))^N\right)\left(1-F_{\gamma_{U_n,2}}(\delta)\right).\end{aligned} \tag{12}$$

The final expression of $F_{\gamma_{1,U_n}}(\psi)$ is derived in Lemma 2.

Lemma 2: Closed-form expression of $F_{\gamma_{1,U_n}}(\psi)$ is given by

$$\begin{aligned}F_{\gamma_{1,U_n}}(\psi) &= \frac{1}{\sqrt{\pi}}\sum_{q=0}^{Q}w_q\Bigg[1-\frac{1}{\Gamma\left[m_{1,U_n}\right]}\sum_{i=0}^{m_{B,1}-1}\frac{(-1)^\tau}{\tau!}\left(\frac{m_{1,U_n}}{\mathcal{A}_{1,U_n}}\right)^{m_{1,U_n}}\\ &\quad\times\left(\frac{m_{B,1}B_3\psi}{\mathcal{A}_{B,1}(B_1-\psi B_2)}\right)^{\tau}\left(\frac{m_{B,1}\mathcal{A}_{1,U_n}B_3\psi}{\mathcal{A}_{B,1}m_{1,U_n}(B_1-\psi B_2)}\right)^{\frac{m_{1,U_n}-\tau}{2}}\\ &\quad\times 2\mathcal{K}_{m_{1,U_n}-\tau}\left(2\sqrt{\frac{m_{B,1}}{\mathcal{A}_{B,1}}\frac{m_{1,U_n}}{\mathcal{A}_{1,U_n}}\frac{B_3\psi}{(B_1-\psi B_2)}}\right)\Bigg].\end{aligned} \tag{13}$$

Proof: Please refer Appendix B for proof of Lemma 2. ■

By utilizing (7) and then performing some mathematical simplifying operations, similar to Appendix B, we can represent $F_{\gamma_{U_n,2}}(\delta)=\Pr[|h_{U_n,2}|^2<\Theta]$, where $\Theta=(\delta/[A_1-A_2\delta])$. The PDF of $|h_{U_n,2}|^2$ can be derived by following similar approach as (2) and substituted in the CDF expression, i.e., $F_{\gamma_{U_n,2}}(\delta)=\int_0^{\Theta}f_{|h_{U_n,2}|^2}(\delta)dz$. Therefore, $F_{\gamma_{U_n,2}}(\delta)$ can be obtained by leveraging [49, eq. (3.381.1)] and is given as

$$F_{\gamma_{U_n,2}}(\delta)=\frac{1}{\sqrt{\pi}}\sum_{c=0}^{C}\frac{w_c}{\Gamma\left[m_{U_n,2}\right]}\Upsilon\left[m_{U_n,2},\Theta\frac{m_{U_n,2}}{\mathcal{A}_{U_n,2}}\right]. \tag{14}$$

2) Intercept Probability: The IP is an event that occurs when the channel capacity of the eavesdropper ($\mathcal{C}_{n,e}$) surpasses the predefined target rate ($\mathcal{R}_{\text{th}}$) [25], [47]. It is mathematically expressed as

$$\mathcal{P}_{\text{int}}=\Pr\left[\mathcal{C}_{U_n,E}>\mathcal{R}_{\text{th}}\right]=1-F_{\gamma_{U_n,E}}(\varphi). \tag{15}$$

We can rewrite $F_{\gamma_{U_n,E}}(\varphi)$ by using (8) as $F_{\gamma_{U_n,E}}(\varphi)=\Pr[|h_{U_n,E}|^2<\Xi]$, where $\Xi=(\varphi/[A_1-A_3\varphi])$. Next, we perform necessary mathematical operations as done to obtain (14) and then applying [49, eq. (3.381.1)] in the acquired results. The final expression of CDF $F_{\gamma_{U_n,E}}(\varphi)$ is given as

$$F_{\gamma_{U_n,E}}(\varphi)=\frac{1}{\sqrt{\pi}}\sum_{g=0}^{G}\frac{w_g}{\Gamma\left[m_{U_n,E}\right]}\Upsilon\left[m_{U_n,E},\Xi\frac{m_{U_n,E}}{\mathcal{A}_{U_n,E}}\right] \tag{16}$$

where $\mathcal{A}_{U_n,E}=10^{(\sqrt{2}\delta_{U_n,E}t_g+\mu_{U_n,E})/10}$ and w_g yields the Gauss-Hermite quadrature's weights.

C. Effective Secrecy Throughput

The SRT analysis, which is performed by obtaining the separate expressions of OP and IP, is insufficient to capture the correlation between the reliability and security aspects of the network [28], [29]. In this direction, EST serves as a crucial metric that offers a holistic characterization of OP and SOP of the system [26]. The final expression for OP is given by (12), whereas we use a well-known Wyner's wiretap code theory to derive an alternative expression for SOP commonly used for evaluating EST and SRT [27], [31]. According to this theory, a secrecy outage occurs when the channel capacity of the wiretap exceeds the redundancy rate of the wiretap code [22], [24], [25]. This is mathematically given as

$$\mathcal{P}_{\text{sop}}^{\text{Wyn}}=\Pr\left[\mathcal{C}_{U_n,E}>(\mathcal{R}_{\text{th}}-\mathcal{R}_{\text{s}})\right]=1-F_{\gamma_{U_n,E}}(\varsigma) \tag{17}$$

where, $(\mathcal{R}_{\text{th}}-\mathcal{R}_{\text{s}})$ is known as the redundancy rate. Further, we can obtain $F_{\gamma_{U_n,E}}(\varsigma)=\Pr[|h_{U_n,E}|^2<\chi]$ by using (8), where $\chi=(\varsigma/[A_1-A_3\varsigma])$. Followed by performing necessary simplifications as done to acquire (14), we can obtain the CDF of $F_{\gamma_{U_n,E}}(\varsigma)$ by the aid of [49, eq. (3.381.1)]. By applying the necessary simplifications, similar to (14), we can obtain the CDF $F_{\gamma_{U_n,E}}(\varsigma)$ with the help of [49, eq. (3.381.1)] as

$$F_{\gamma_{U_n,E}}(\varsigma)=\frac{1}{\sqrt{\pi}}\sum_{o=0}^{O}\frac{w_o}{\Gamma\left[m_{U_n,E}\right]}\Upsilon\left[m_{U_n,E},\chi\frac{m_{U_n,E}}{\mathcal{A}_{U_n,E}}\right] \tag{18}$$

where $\mathcal{A}_{n,e}=10^{(\sqrt{2}\delta_{n,e}t_o+\mu_{n,e})/10}$ and w_o yields the Gauss-Hermite quadrature's weights. Additionally, it is evident from (4), (7), and (8) that the channels included in the expressions of $\mathcal{P}_{\text{out}}$ and $\mathcal{P}_{\text{sop}}^{\text{Wyn}}$ are independent of each other. This results in an important conclusion that $\mathcal{P}_{\text{out}}$ is also independent of $\mathcal{P}_{\text{sop}}^{\text{Wyn}}$. Moreover, in the considered system model, the eavesdropper can wiretap only in the second IT phase. So, $\mathcal{P}_{\text{sop}}^{\text{Wyn}}=0$ in the first IT phase. Based on these discussions, EST is mathematically given by

$$\mathcal{T}_{\text{eff}}^{\text{sec}}=(1-\beta)(1-\alpha)(1-\mathcal{P}_{\text{out}})\left(1-\mathcal{P}_{\text{sop}}^{\text{Wyn}}\right)\mathcal{R}_{\text{s}}. \tag{19}$$

It can be observed from (19) that the network's throughput depends upon the outage event $\mathcal{P}_{\text{out}}$ and the secrecy outage event $\mathcal{P}_{\text{sop}}^{\text{Wyn}}$. Therefore, the main focus of the work is to

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maximize EST of the network while meeting the specified constraints on $\mathcal{P}_{\text{out}}$ and $\mathcal{P}_{\text{sop}}^{\text{Wyn}}$. In the next section, we formulate the EST maximization problem by utilizing the final expressions of EST obtained in (19).

IV. Problem Formulation and Proposed Algorithm

In this section, we formulate the EST maximization problem by defining the constraints and optimization variables. Followed by providing an efficient algorithm that maximizes the EST of the considered system.

A. Problem Formulation

We formulate the optimization problem for maximizing the EST by satisfying the constraints on the system parameters, U's locations, and transmit powers. In particular, we optimize the system parameters (α, β, ϵ), U_n's locations $(x_{U_n}, y_{U_n}, H_{U_n})$. Specifically, $\mathcal{D}$ is the set of optimizing variables, i.e., N, the tuples $(x_{U_n}, y_{U_n}, H_{U_n})$ $\forall n$, α, β, ϵ, P_1, and P_{U_n}. The maximum transmit powers of IoD_1 and U_n are denoted as $P_1^{\max}$ and $P_{U_n}^{\max}$, respectively. Given the final expression of EST in (19), we formulate the EST maximization problem (P_1) as

$$
\begin{aligned}
(\text{P1}): \max_{\mathcal{D}} \quad & \mathcal{T}_{\text{eff}}^{\text{sec}} && (20a)\\
\textbf{s.t.} \quad & \mathcal{P}_{\text{out}} \le \gamma_1 && (20b)\\
& \mathcal{P}_{\text{sop}}^{\text{Wyn}} \le \gamma_2 && (20c)\\
& 0 \le P_1 \le P_1^{\max} && (20d)\\
& 0 \le P_{U_n} \le P_{U_n}^{\max} && (20e)\\
& 0 < \alpha < 1, \quad 0 < \epsilon \le 1, \quad 0 < \beta < 1 && (20f)\\
& x_{U_n}^{\min} \le x_{U_n} \le x_{U_n}^{\max}, \ y_{U_n}^{\min} \le y_{U_n} \le y_{U_n}^{\max} \\
& H^{\min} \le H_{U_n} \le H^{\max} && (20g)
\end{aligned}
$$

$(\cdot)^{\min}$ and $(\cdot)^{\max}$ denotes the minimum and maximum values of the variable. The constraints of (P1) given in (20a) are described as (20b) ensure the reliability of the network by accounting for successful transmissions between the IoDs via U_n. In contrast, (20c) guarantees the confidentiality of the information by preventing interception. Additionally, (20d) and (20e) impose limits on the transmit powers of IoD_1 and U_n. Finally, the constraints on the system parameters α, β, and ϵ are defined in (20g), respectively. The selected ranges for α, β, and ϵ are based on practical considerations and have been adopted in [18], [19], [30], [33], and [34]. These constraints are carefully chosen to ensure an optimal time allocation for EH, IT phases, and power allocation for AN injection. Whereas the constraint defined in (20g) corresponds to restrictions on the positions of U_R. Initially, we can observe that the problem outlined (P_1) given in (20a) is a complex nonlinear optimization problem with multiple local maxima. It is defined over a high-dimensional continuous search space, which includes optimizing variables, such as N, $(x_{U_n}, y_{U_n}, H_{U_n})$, α, β, ϵ, P_1, and P_{U_n}, subject to various constraints. To solve the EST maximization problem, we combine the PSO algorithm with the CGA in the next section.

B. Evolutionary and Swarm Intelligence-Based Algorithm

In this section, we combine the fast greedy convergence of PSO and the effective exploration capability of GA and propose a hybrid evolutionary and swarm intelligence-based algorithm to solve (P1). The CGA algorithm is an enhanced version of the GA, which can efficiently handle a large number of continuous variables. First, we provide a detailed construction of the input for our proposed algorithm and outline its structure. Specifically, the chromosome is constructed as a vector of real values in CGA, where a ψth chromosome in the τth generation is defined as

$$\mathcal{D}(\tau)_{\psi} = \Big[N, \big[x_{U_n}, y_{U_n}, H_{U_n}\big]_{N\times 3}, P_1, P_{U_n}, \alpha, \beta, \epsilon\Big] \quad (21)$$

where N represents the number of UAVs, and dimensions of the matrix $[x_{U_n}, y_{U_n}, H_{U_n}]_{N\times 3}$ change depending on N. This implies that when a new solution candidate uses more UAVs than the previous one, the additional UAV positions must be added to the chromosome and optimized. For the viable heuristic search, we adjust the candidate chromosome $\mathcal{D}(\tau)_{\psi}$ by setting the N dimension of the matrix $[x_{U_n}, y_{U_n}, H_{U_n}]_{N\times 3}$ to its upper bound, $N = N^{\max}$. Therefore, (21) can be modified as

$$\mathcal{D}(\tau)_{\psi} = \Big[N, \big[x_{U_n}, y_{U_n}, H_{U_n}\big]_{N^{\max}\times 3}, P_1, P_{U_n}, \alpha, \beta, \epsilon\Big] \quad (22)$$

where $N^{\max}$ is the maximum number of UAVs. Importantly, if $N < N^{\max}$, only the first N rows of the matrix $x_{U_n}, y_{U_n}, H_{U_n}]_{N^{\max}\times 3}$ are relevant for that solution candidate. The algorithm starts with randomly initialized chromosomes in the population $\mathcal{D}(0)_{\psi}$. Each chromosome is evaluated using the objective function in (20a). Promising chromosomes are selected for reproduction during the selection phase to maintain the population size (Λ). In the crossover step, most chromosomes are paired and recombined with a crossover rate (R_{co}) to generate offspring. Specifically, when the hybrid PSO-CGA algorithm selects a pair of chromosomes $\mathcal{D}(\tau)_{\breve{p}}$ and $\mathcal{D}(\tau)_{\breve{q}}$ which creates two new candidate chromosomes, denoted as $\mathcal{D}(\tau+1)_{\breve{p}}$ and $\mathcal{D}(\tau+1)_{\breve{q}}$ and obtained as

$$\mathcal{D}(\tau+1)_{\breve{p}} = \mathcal{U}_1\mathcal{D}(\tau)_{\breve{p}} + \mathcal{U}_2\mathcal{D}(\tau)_{\breve{q}} + \mathcal{U}_3\mathcal{D}(\tau)_{\text{best}} \quad (23)$$

$$\mathcal{D}(\tau+1)_{\breve{q}} = \mathcal{U}_2\mathcal{D}(\tau)_{\breve{p}} + \mathcal{U}_1\mathcal{D}(\tau)_{\breve{q}} + \mathcal{U}_3\mathcal{D}(\tau)_{\text{best}} \quad (24)$$

where $\mathcal{U}_k$ are uniform random values such that $\mathcal{U}_k > 0$ and $\sum_{k=1}^{3}\mathcal{U}_k = 1$. The core concept of the proposed methodology lies in its crossover mechanism, which incorporates not just two parents, as in traditional GA approaches, but three—two selected parents and the best individual from all previous generations. This design enables a dual search strategy: 1) a local exploration around the parents and 2) a greedy search directed toward the global best candidate. By leveraging this hybrid mechanism, the approach effectively combines the strengths of CGA's diversity and PSO's rapid convergence, ensuring that offspring are guided toward optimal solutions. Despite the added complexity of incorporating the influence of $\mathcal{D}(\tau)_{\text{best}}$ in the crossover, the computational overhead remains minimal, requiring only a few additional multiplications. Thus, the algorithm maintains computational efficiency similar to CGA while offering enhanced performance, particularly in escaping local maxima and accelerating convergence. The mutation phase

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introduces further diversity by modifying chromosomes at a low mutation rate (R_m). This process involves adding random values sampled from a Gaussian distribution with a mean of 0 and a standard deviation of 1/20 relative to the upper-bound entry length. This step is essential to counteract the tendency of chromosomes to become homogeneous over successive generations. The algorithm iteratively refines the population through a loop of selection, crossover, and mutation. At the end of iterations, the optimized chromosome representing the best solution is returned, which is given as

$$\mathcal{D}^* = \left[N^*, \left[x^*_{U_n}, y^*_{U_n}, H^*_{U_n}\right]_{N^{\max}\times 3}, P^*_1, P^*_{U_n}, \alpha^*, \beta^*, \epsilon^*\right]. \quad (25)$$

This methodology balances exploration and exploitation, making it robust for complex optimization tasks.

C. Penalty, Convergence, and Computational Complexity

First, we simplify the constrained optimization problem by modifying the objective function to introduce penalties for constraint violations described in (20b)–(20f). The fitness value for an individual solution candidate is expressed as

$$\begin{aligned}\text{Fitness}(\mathcal{D}_\psi) = &\left(\mathcal{T}^{\text{sec}}_{\text{eff}}\right) - \Omega_1 \max\left(0, P_{U_n} - P^{\max}_{U_n}\right)\\ &- \Omega_2 \max\left(0, P_1 - P^{\max}_1\right) - \Omega_3 \max(0, \mathcal{P}_{\text{out}} - \gamma_1)\\ &- \Omega_4 \max\left(0, \mathcal{P}^{\text{Wyn}}_{\text{sop}} - \gamma_2\right), \end{aligned} \quad (26)$$

where $\Omega_i > 0\ \forall i \in \{1, 2, 3, 4\}$ ensure that the penalties are applied appropriately.[2]

1) Convergence Analysis: Now, we analyze the convergence behavior of the proposed hybrid algorithm by examining its key stages, i.e., selection, crossover, and mutation. During the selection stage, the probability $\mathcal{P}_{(\tau+1)}(\mathcal{D}_{\check{p}})$ of selecting an individual $\mathcal{D}_{\check{p}}$ for reproduction in generation $\tau + 1$ is proportional to its fitness $\mathcal{F}_{\check{p}}$ in generation τ, given as

$$\mathcal{P}_{(\tau+1)}\left(\mathcal{D}_{\check{p}}\right) = \frac{\mathcal{F}_{\check{p}}}{\sum_{\check{q}} \mathcal{F}_{\check{q}}}. \quad (27)$$

This mechanism favors individuals with higher fitness, causing their probabilities to grow exponentially over generations while the likelihood of selecting low-fitness individuals diminishes. Assuming a uniform initial fitness distribution and a maximum fitness M, the average fitness in generation τ, denoted by $\bar{\mathcal{F}}_\tau$, is given as

$$\bar{\mathcal{F}}_\tau = \frac{\tau + 1}{\tau + 2} M. \quad (28)$$

This indicates that the average fitness approaches M at a rate of $M/(\tau + 2)$, demonstrating the convergence trend. In the crossover stage, variability is introduced by combining genetic material from selected parents. Unlike traditional GAs, our hybrid approach generates offspring using three parents, i.e., $\mathcal{D}(\tau)_{\check{p}}$, $\mathcal{D}(\tau)_{\check{q}}$, and the current best individual $\mathcal{D}(\tau)_{\text{best}}$. The offspring, computed in (23) and (24), includes $\mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3 > 0$ with $\mathcal{U}_1 + \mathcal{U}_2 + \mathcal{U}_3 = 1$. This triangular combination allows exploration within the convex hull formed by the three parents, enhancing the local search capability and promoting diversity. Lastly, the mutation stage further introduces variability by adding Gaussian noise to selected individuals, ensuring exploration beyond the current search space. This step helps maintain genetic diversity and prevents premature convergence. Overall, the hybrid approach leverages the greedy and uphill nature of incorporating $\mathcal{D}_{\text{best}}$ for local searches, improving the likelihood of generating high-fitness offspring. The mean fitness curve exhibits progressive improvement and occasional fluctuations due to mutation effects, leading to faster and more robust convergence.

[2]The proposed CGA-PSO algorithm can incorporate additional reward and penalty functions to address more practical scenarios. However, addressing the constraints, such as peak and average power limits, information and energy causality, and UAV collision avoidance will inevitably expand the search space and increase the computational complexity of the optimization process.

Algorithm 1: Proposed Algorithm to Maximize EST

1) **Initialize:** Set population size Λ, crossover rate R_{co}, mutation rate R_m, maximum iterations τ^m, and initialize $\tau = 0$.
2) **Procedure:** Generate the initial population $\mathcal{D}(\tau)_\psi, \forall\psi = 1, \ldots, \Lambda$.
3) **for** $\tau = 1$ to τ^m, **do:**
4) **for** $\psi = 1$ to Λ, **do:**
5) Evaluate the fitness of ψ-th chromosome: $\mathcal{F}^{(\tau)}_\psi = \text{Fitness}(\mathcal{D}^{(\tau)}_\psi)$ using (26).
end.
6) Generate new chromosomes through reproduction based on their fitness values.
7) Perform crossover to create offspring by merging current best individual with the parents using R_{co}.
8) Introduce mutations in the offspring with R_m.
9) Form the new population.
end
10) **Output:** $\mathcal{D}^*$.

2) Computational Complexity: The computational complexity of the proposed algorithm, as shown in Algorithm 1, is dominated by the for loops (steps 3 and 4). The first loop (step 3) creates each generation, whereas the subsequent loop (step 4) processes each individual in the population for each generation. Since the fitness function depends upon a feed-forward loop, resulting in an overall complexity of $O(\Lambda)$ for the entire population. The crossover and mutation operations require an additional loop with a complexity of $O(P_{\text{co}}\Lambda + P_m\Lambda)$. The roulette wheel selection (step 7) has a complexity of $O(\Lambda)$ for forming the selection array, and selecting the individual has a complexity of $O(\ln \Lambda)$. Therefore, the total complexity of Algorithm 1 is obtained as $O(\Lambda\tau^m) + O((P_{\text{co}}\Lambda + P_m\Lambda)\tau^m) + O(\Lambda\tau^m) + O((\Lambda \ln \Lambda)\tau^m)$, where τ^m is the maximum number of iterations.

V. Numerical Results

In this section, we present the numerical and simulation results for the analysis of SOP, SRT, and EST, aiming to validate the precision of the analytical formulations. Throughout the numerical results, $P_B = P_{U_n} = P$ and express the transmit SNR as P/N_0. The parameters α, β, and ϵ are assigned specific values, i.e., $\alpha = 0.3$, $\beta = 0.4$, and $\epsilon = 0.4$. Furthermore, we set carrier frequency $f_c = 700$ MHz. The

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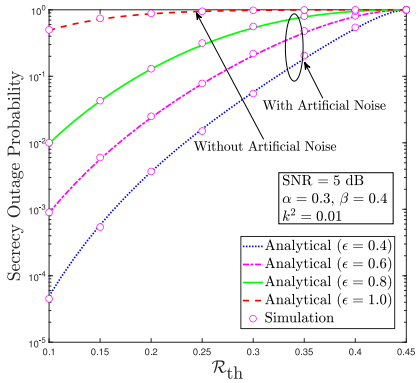

Fig. 2. SOP versus $\mathcal{R}_{\text{th}}$ with different ϵ values.

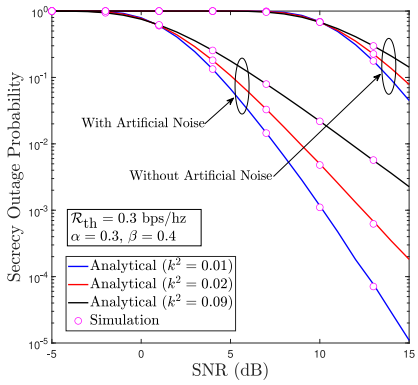

Fig. 3. SOP versus SNR for different HI levels and ϵ values.

terrestrial nodes i.e., (B), IoD_1, IoD_2, and Eve are located at $(0, 50, 0)$, $(50, 50, 0)$, $(150, 50, 0)$, and $(180, 50, 0)$. Unless specified in the optimization results, we set the coordinates of UAVs as $(70, 50, 50)$, $(80, 50, 50)$, and $(90, 50, 50)$. Whereas, for optimizing the UAV's position, we take $x_{U_n}^{\min} = 0$, $x_{U_n}^{\max} = 180$, $y_{U_n}^{\min} = 25$, $y_{U_n}^{\max} = 75$, $H^{\min} = 0$, and $H^{\max} = 100$ m. The target data rate is specified as $\mathcal{R}_{th} = 0.25$ bps/Hz, and the confidential data rate is defined as $\mathcal{R}_{\text{s}} = 0.1$ bps/Hz. Considering the dense urban scenario, the following parameters are considered, i.e., $a = 12.081$, $b = 0.1139$, $(c_1, d_1) = (9.64, 0.04)$, $(c_2, d_2) = (30.83, 0.04)$, and $(\psi_{\text{LoS}}, \psi_{\text{NLoS}}) = (1, 20)$. The fading severity parameter is set as $m_{i,j} = m = 2$ for simplicity. The network configuration is considered for single and multiple UAV-assisted IoD scenarios, denoted as $N = 1, 3$, whereas $N^{\max} = 5$. The energy efficiency parameter is set to $\eta_1 = 0.8$, and HI level, $k^2 = 0.01$. The Gauss-Hermite quadrature approximation coefficients, i.e., $L = G = Q = O = C = J = 15$ in the final expressions of SOP, IP, OP, and EST, given in (10), (13), (14), (16), and (18), respectively. We have performed Monte-Carlo simulations for 10^5 iterations to ensure accurate simulation results.

A. Secrecy Outage Probability Analysis

Figs. 2–7 show good agreement between the simulated results and the analytical results obtained using SOP expression derived in (10) under various parameter settings. First,

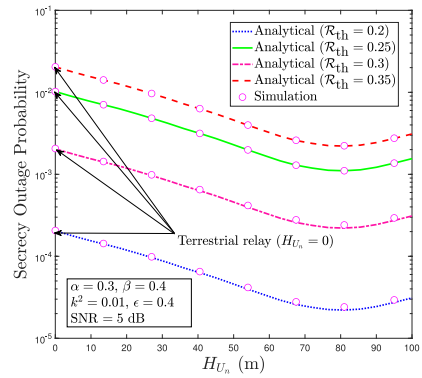

Fig. 4. SOP versus UAV's height with different $\mathcal{R}_{\text{th}}$.

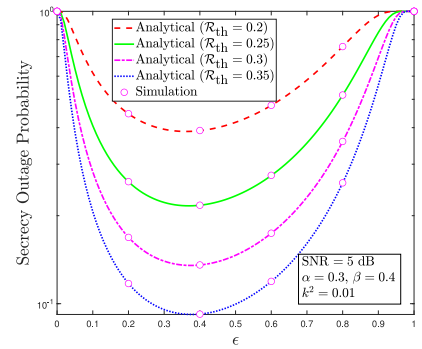

Fig. 5. SOP versus ϵ with different $\mathcal{R}_{\text{th}}$.

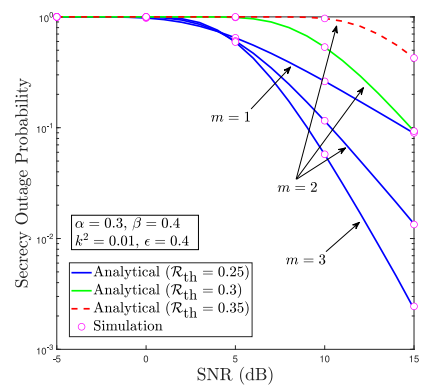

Fig. 6. SOP versus SNR for different $\mathcal{R}_{\text{th}}$ and m.

we demonstrate the impact of considering the AN injection technique and compare it with a baseline method that does not use AN injection. To model the system without AN injection, we set $\epsilon = 1$ in (6), i.e., all the power of the selected UAV is allotted to IT, and no power is assigned to AN. Fig. 2 shows the variation of SOP with different values of $\mathcal{R}_{\text{th}}$ and ϵ. It is evident from Fig. 2 that, including AN in the UAV's transmitted signal significantly degrades the eavesdropper's SNDR, thereby reducing the SOP and enhancing the overall system's security performance.

The curves in Fig. 2 also indicate that SOP increases as $\mathcal{R}_{\text{th}}$ increases from $0.1 - 0.45$ bps/Hz for the given set

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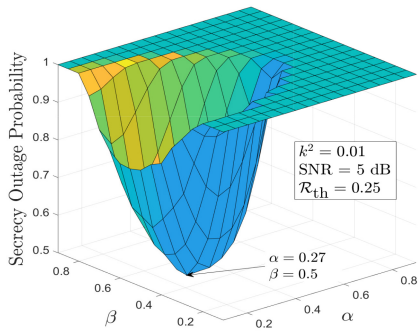

Fig. 7. SOP versus α and β.

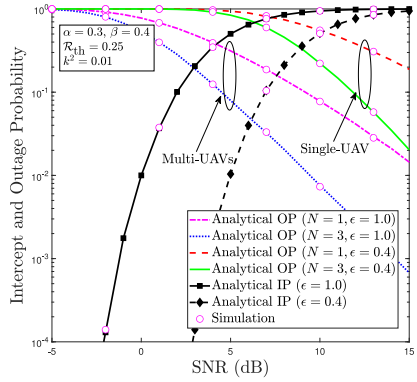

Fig. 8. IP and OP versus transmit SNR with different N and ϵ values.

of parameters. This behavior occurs because the achievable instantaneous rate with the fixed transmit SNR tends to fall below the increasing target rate, resulting in degraded secrecy performance. Moreover, it is evident from the curves that SOP performance for the lower ϵ value, i.e., below $\epsilon = 0.4$, is considerable as compared to higher ϵ values. Fig. 3 illustrates the SOP versus SNR for various HI levels ($k^2 = 0.01, 0.02, 0.09$) and different ϵ values. The results show that increasing the SNR values reduces the SOP, demonstrating enhanced secrecy performance at higher SNR levels. This reduction is impacted by the level of HIs. Larger values of k^2 lead to consistently higher SOP, signifying that more severe HIs degrade the system's secrecy. Additionally, including AN injection ($\epsilon = 0.4$) improves secrecy performance compared to the case without AN ($\epsilon = 1$). This improvement is attributed to the AN's ability to reduce the eavesdropper's capacity to intercept the communication effectively.

Fig. 4 provides valuable insights about the SOP curves versus UAV's varying heights for different values of $\mathcal{R}_{\text{th}}$. It is apparent from the curves that the UAV relay performs better than the terrestrial relay up to a certain height, beyond which the SOP performance degrades. Initially, when H_{U_n} increases, the LoS path between the UAV and IoD_2 improves, which results in reduced path loss and stronger received signal strength at the legitimate receiver, thus reducing the SOP. However, as the UAV continues to fly beyond the particular height, the eavesdropper may intercept the signal, thus increasing the SOP. Therefore, the UAV's flying altitude should be optimally designed to improve the secrecy performance. For instance, the SOP value for $\mathcal{R}_{\text{th}} = 0.2$ bps/Hz is 0.02045 when $H_{U_n} = 0$, but it significantly improves to a value of 0.00224 at $H_{U_n} = 79.5$ m. But, beyond $H_{U_n} = 79.5$ m, SOP performance starts degrading.

Fig. 5 illustrates the SOP curves versus ϵ for various $\mathcal{R}_{\text{th}}$ values. These curves show that the SOP is at its lowest for the mid-range of ϵ and then experiences a notable rise for lower and higher ϵ values. Specifically, lower ϵ values reflect that the power assigned to IT is very low, thereby resulting in increased SOP. Conversely, when ϵ values are higher, the amount of power assigned to AN decreases, leading to poor SOP performance. Hence, it can be concluded that the optimal ϵ value for achieving minimum SOP can be obtained from the curves shown in Fig. 5. Specifically, the optimal ϵ value with $\mathcal{R}_{\text{th}} = 0.35$ bps/Hz and SNR = 5 dB is 0.39, whereas it is 0.32 with $\mathcal{R}_{\text{th}} = 0.2$ bps/Hz. Likewise, Fig. 6 illustrates the SOP performance for different values of $\mathcal{R}_{\text{th}}$ and m with varying transmit SNR. The curves indicate that the SOP at IoD_2 decreases as the transmit SNR increases for a fixed $\mathcal{R}_{\text{th}}$ value. Nevertheless, the SOP performance deteriorates with increasing $\mathcal{R}_{\text{th}}$ values, which is also evident from Fig. 2. This behavior is evident because the instantaneous achievable secrecy rate tends to fall below the increasing $\mathcal{R}_{\text{th}}$ value at the fixed transmit SNR. Additionally, it is obvious that increasing the value of $m \in 1, 2, 3$ improves the SOP performance.

We use 3-D plot in Fig. 7 to show the SOP performance for different α and β values. Since the time allocation for WPT and overall IT between the IoDs is determined by these parameters, it is essential to understand their combined impact on the system's performance and identify the optimal values to achieve the desired performance. The convex shape of the SOP plot indicates the presence of an optimal combination of α and β. Specifically, minimum SOP corresponding to $\mathcal{R}_{\text{th}} = 0.25$ bps/Hz and transmit SNR = 5 dB, can be attained at $\alpha = 0.27$ and $\beta = 0.5$.

B. Analyzing the Security-Reliability Tradeoff

SRT analysis enables the system designers to understand the tradeoffs between the security and reliability aspects of the system, which can be improved by optimization and resource allocation of system parameters. Figs. 8–10 provide valuable insights into the SRT of the considered system. We first compare the proposed AN injection and multi-UAV relaying with the conventional IT method and single UAV relaying. The curves in Fig. 8 depict SRT performance, i.e., IP and OP as functions of transmit SNR for different N and ϵ values. Interestingly, the outage performance improves with increasing transmit SNR and the number of UAV relays, which is due to the fact that a higher number of UAV relays can provide a better diversity order, thereby enhancing the outage performance.

It is essential to note that higher transmit SNR enhances the channel capacity of the eavesdropper's wiretap link, subsequently increasing the IP. This presents a tradeoff between security and reliability when transmitting information in the presence of an eavesdropper. However, it is observed from Figs. 8 and 9 that the SRT can be significantly improved

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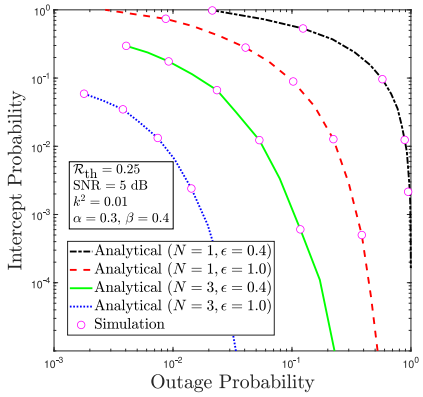

Fig. 9. Analyzing SRT with IP versus OP curves.

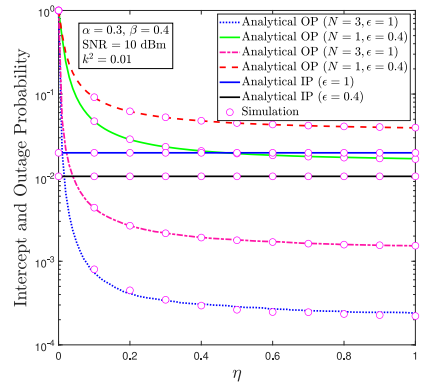

Fig. 10. IP and OP versus energy conversion efficiency (η) with N and ϵ.

by employing the AN injection and relay selection techniques together. Moreover, with multi-UAV relaying and AN injection, the system can improve secrecy and reliability by adjusting system parameters. For instance, with $\epsilon = 1$, the OP and IP values obtained in Fig. 8 for a single UAV relay ($N = 1$) are 0.78654, which reduced to 0.14983 when multiple UAV relaying ($N = 3$) has been employed. Whereas, with $\epsilon = 0.4$, the values of OP and IP obtained for a single UAV relay are 0.67981, significantly reduced to 0.051138 for the multiple UAV scenario. This validates the fact that multi-UAV relaying and AN injection can enhance the reliability and information secrecy of the network.

Fig. 10 demonstrates the network's IP and OP variations with respect to η for different number of UAVs and ϵ values. As η increases, the OP decreases in all scenarios, meaning communication becomes more reliable because higher energy conversion efficiency provides more power for transmitting signals. This decrease in OP is more noticeable in the multi-UAV scenario ($N = 3$) compared to the single-UAV scenario ($N = 1$), as multiple UAVs offer better spatial diversity. Nevertheless, the OP is higher when $\epsilon = 0.4$ compared to $\epsilon = 1$ because, with $\epsilon = 0.4$, less power is allocated to data transfer, which degrades the outage performance. In contrast, when $\epsilon = 1$, the UAV's full power is utilized for data transfer, significantly reducing the OP. Whereas the IP remains constant for all values of η, i.e., the energy conversion efficiency does not affect the eavesdropper's ability to intercept the signal [can also be inferred from (16)]. However, the AN injection scheme ($\epsilon = 0.4$) reduces the IP compared to the no AN injection case ($\epsilon = 1$) because the AN degrades the eavesdropper's wiretap link, limiting its interception capabilities.

C. Effective Secrecy Throughput Analysis

Figs. 11 and 12 depict the EST curves under various system parameters, demonstrating the accuracy of both simulated results and analytical EST expressions. Fig. 11 highlights the impact of $\mathcal{R}_s$ variations on the EST for different values of ϵ and N. It can be observed that the EST can be concurrently improved by increasing the number of UAVs in the network for lower-mid transmit SNR regime. However, as transmit SNR becomes large, the impact of N on EST tends to converge to a similar curve. This behavior arises from the fact that increasing

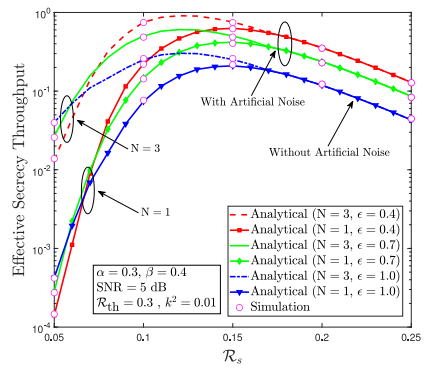

Fig. 11. EST versus $\mathcal{R}_s$ for different N and ϵ values.

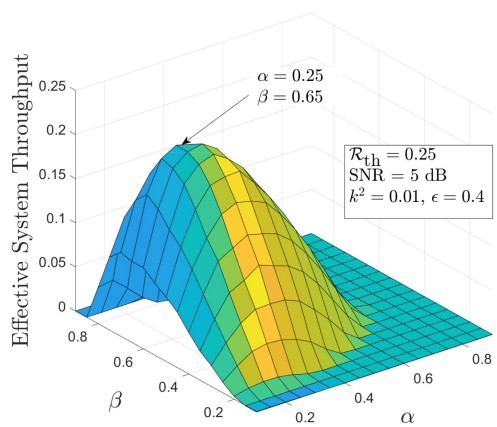

Fig. 12. EST versus α and β.

N leads to improved reliability performance. Nevertheless, the security is not related to the increase in N. Moreover, the lower ϵ shows superior EST performance compared to higher ϵ values in the low SNR region. With the decrease in ϵ, the power corresponding to the IT reduces while the jamming power increases, thus reducing the SOP.

Whereas, after attaining the maximum EST in the mid-SNR regime, it starts degrading due to the rise in the OP with the increase in ϵ from 0.4–1. This EST behavior is evident due to its dependency on OP and SOP, thus providing valuable insight into achievable EST for the considered set of parameters. As

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TABLE II
COMPARISON OF PROPOSED AND BASELINE ALGORITHMS

Algorithm	Number of Generations (95% convergence)	Run time (sec)	Optimal EST	Computational Complexity
CGA	14.7	1.8137	1.892	$O\left(\tau^m \Lambda \left(C_f + \ln \Lambda + P_{\text{co}} C_{\text{cro}} + P_m C_{\text{mut}}\right)\right)$
PSO	3.9	1.3913	2.083	$O\left(\tau^m \Lambda (C_f + G)\right)$
Hybrid PSO [35]	4.2	1.7974	2.333	$O(\tau^m \Lambda (C_f + G + C_{LS} + P_{\text{co}} C_{\text{cro}} + P_m C_{\text{mut}}))$
GA-PSO [36]	15.2	2.3146	2.232	$O\left(\tau^m \Lambda \left(C_f + P_{\text{co}} C_{\text{pmx}} + P_m C_{\text{mut}} + G + \ln \Lambda\right)\right)$
Proposed work	14.30	1.7958	2.417	$O(\Lambda \tau^m) + O((P_{\text{co}} \Lambda + P_m \Lambda)\tau^m) + O(\Lambda \tau^m) + O((\Lambda \ln \Lambda)\tau^m)$

shown in Fig. 11, for $\mathcal{R}_s = 0.1$, a single-UAV system without AN injection achieves an EST as 0.7468. In contrast, the proposed method achieves an EST as 0.0765, indicating an improvement of nearly 10 times.

Interestingly, Fig. 12 shows the 3-D plot of EST performance versus different values of α and β. The parameters α and β controls and governs the time allocation for WPT and IT between the IoDs. These parameters play a critical role in determining the system's overall performance by balancing EH and data transmission efficiency. Analyzing their joint impact is essential to identify optimal values that maximize the EST under given constraints. The results indicate that there exists an optimal combination of α and β that leads to peak system performance. Specifically, when $\alpha = 0.25$ and $\beta = 0.65$, the system achieves maximum EST. This optimal point reflects an effective tradeoff between dedicating sufficient time for EH to sustain the IoDs and allocating adequate time for reliable data transmission.

D. Results With Optimized System Parameters

This section compares the performance of proposed algorithm with the other baseline and existing algorithms from the literature. Specifically, the proposed hybrid PSO-CGA method is compared with different algorithms, i.e., CGA, PSO, and hybrid methods in [35] and [36]. The total computational complexity of the CGA is $O(\tau^m \Lambda (C_f + \ln \Lambda + P_{\text{co}} C_{\text{cro}} + P_m C_{\text{mut}}))$, where C_f represents the complexity of the fitness function, P_{co} is the crossover probability, C_{cro} is the crossover complexity, P_m is the mutation probability, and C_{mut} is the mutation complexity. The total computational complexity of the PSO algorithm is $O(\tau^m \Lambda (C_f + G))$, where C_f is the fitness evaluation complexity and G is the number of generations. The total computational complexity of the hybrid PSO algorithm described in [35] is $O(\tau^m \Lambda (C_f + G + C_{LS} + P_{\text{co}} C_{\text{cro}} + P_m C_{\text{mut}}))$, where C_{ls} is the complexity of local search. Finally, the GA-PSO algorithm proposed in [36] has a computational complexity of $O(\tau^m \Lambda (C_f + P_{\text{co}} C_{\text{pmx}} + P_m C_{\text{mut}} + G + \ln \Lambda))$, where C_{pmx} is the complexity of the partial matching crossover operation. The convergence behavior of the proposed and other baseline algorithms for 40 generations is depicted in Fig. 13, and their comparison is summarized in Table II, which emphasizes the superiority of the proposed PSO-CGA-based hybrid algorithm.

The comparative analysis of the algorithms is performed based on the required number of generations to achieve 95% of convergence, real runtime, and obtained optimal EST values. The PSO method demonstrates the fastest convergence among the baseline algorithms, requiring only 3.9 generations to achieve 95% convergence. However, this speed comes at the cost of suboptimal results, with an Optimal EST of 2.083. In contrast, CGA achieves a convergence rate of 14.7 generations, comparable to the proposed method (14.3 generations). Although the number of generations measures the convergence speed, actual run time is also crucial, as different methods involve varying computational efforts. Despite the similarity in convergence speed, CGA delivers the lowest Optimal EST of 1.892 and a slightly higher runtime of 1.8137 s, making it less competitive.

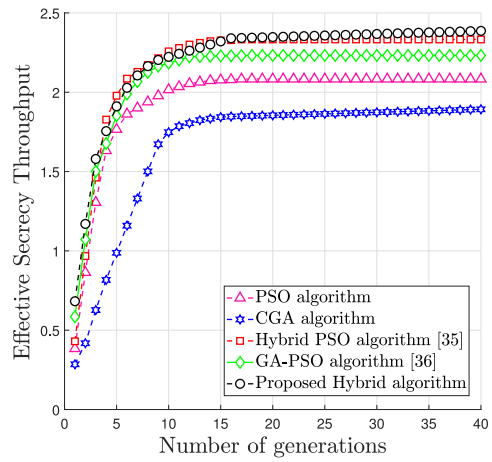

Fig. 13. Convergence curves for the proposed and baseline algorithms.

The hybrid PSO algorithm in [35] balances convergence speed and optimality, converging in 4.2 generations with an Optimal EST of 2.333. On the other hand, the GA-PSO algorithm in [36] demonstrates robust optimization with an Optimal EST of 2.232 but suffers from a significantly longer runtime of 2.3146 s and a slower convergence rate of 15.2 generations, making it less efficient overall. The proposed algorithm outperforms all baseline methods in optimality, achieving the highest Optimal EST of 2.417. Its runtime of 1.7958 s is slightly higher than PSO but lower than baseline algorithms in [35] and [36], highlighting its computational efficiency. These findings suggest that the proposed CGA-PSO-based hybrid algorithm effectively balances convergence speed, runtime, and optimality.

VI. CONCLUSION

This work focused on establishing a secure and reliable UAV-relay link between the energy-constrained IoD pairs, particularly in scenarios where direct communication links

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are severely obstructed or absent. Based on various benefits offered by multiple UAVs, such as enhanced reliability and better performance in terms of communication quality and system robustness, we incorporate multi-UAVs in the considered system. We employed SNDR-based relay selection to improve the link's reliability and AN injection technique to counter the security threat the eavesdropper poses. The investigation incorporates HIs to assess the secrecy performance through SOP and EST, and further SRT analysis is conducted based on IP and OP. The accurate expressions of SOP, IP, and OP are derived considering the composite fading channel model consisting of log-normal shadowing and Nakagami-m small scale fading. Moreover, we formulated the EST maximization problem and optimized the UAV's position and system parameters by considering the outage and secrecy constraints. To address the nonlinear optimization problem, we proposed a CGA-PSO hybrid algorithm and evaluated its performance against baseline algorithms, including individual PSO and GA algorithms and the hybrid algorithms presented in earlier published papers. This work has demonstrated the potential of multi-UAV-enabled secure and reliable communications. There are several promising areas for future research that have yet to be explored. For instance, one can extend the proposed model by incorporating the effects of imperfect CSI and UAV mobility. Additionally, machine learning techniques can be used to optimize UAVs' trajectories for dynamic adaptation in unpredictable scenarios to further enhance system robustness.

Appendix A

We substitute the expressions of $\gamma_{U_n,2}$ and $\gamma_{U_n,E}$ obtained in (7) and (8) into (9) to acquire $\mathcal{P}_{\text{sop}}$. We take $A_1 = (\epsilon P_{U_n}/N_0)$, $A_2 = ([(1-\epsilon)P_{U_n}]/N_0)$, and $A_3 = ([(1-\epsilon+k^2)P_{U_n}]/N_0)$. Moreover, the variables $X = |h_{U_n,2}|^2$ and $Y = |h_{U_n,E}|^2$ follow composite channel model, comprising log-normal and gamma distribution accounting for shadowing and Nakagami-m faded signal amplitude distribution, whose PDFs are given in (2). Therefore, the reformed expression of $\mathcal{P}_{\text{sop}}$ is given as

$$\mathcal{P}_{\text{sop}} = \Pr\left[X < \frac{\phi Y}{1-(\phi A_2 - A_3)Y}\right] = \underbrace{\int_0^{\frac{1}{\phi A_2 - A_3}} f_Y(y)dy}_{I_1} + \underbrace{\int_{\frac{1}{\phi A_2 - A_3}}^{\infty} f_Y(y) \int_0^{\frac{\phi y}{1-(\phi A_2 - A_3)y}} f_X(x) f_Y(y) dx dy}_{I_2}. \quad (29)$$

In (29), I_1 can be evaluated using [49, eqs. (3.381.1)], and expressed as

$$I_1 = \frac{1}{\sqrt{\pi}} \sum_{\varrho=0}^{\mathcal{K}} \frac{w_\varrho}{\Gamma[m_{U_n,E}]} \Upsilon\left[m_{U_n,E}, \frac{m_{U_n,E}}{\phi A_2 - A_3 \mathcal{A}_{U_n,E}}\right]. \quad (30)$$

Subsequently, the PDFs of X and Y are substituted into I_2. By employing [49, eq. (381.3)] and performing a change of variables, I_2 is obtained as

$$\begin{aligned} I_2 = \frac{1}{\pi} \sum_{\kappa=0}^{\mathcal{J}} \sum_{l=0}^{L} \frac{w_\kappa w_l}{\Gamma[m_{U_n,E}]} \Bigg[\Gamma\left(m_{U_n,E}, \frac{m_{U_n,E}}{\phi A_2 - A_3 \mathcal{A}_{U_n,E}}\right) - \frac{1}{p!} \\ \times \sum_{p=0}^{m_{U_n,2}-1} \left(\frac{(-1)\times m_{U_n,2}}{\phi A_2 - A_3 \mathcal{A}_{U_n,2}}\right)^p \left(\frac{m_{U_n,E}}{\mathcal{A}_{U_n,E}} \frac{1}{\phi A_2 - A_3}\right)^{m_{U_n,E}} \\ \times e^{-\left(\frac{1}{\phi A_2 - A_3}\left(\frac{m_{U_n,E}}{\mathcal{A}_{U_n,E}} - \frac{m_{U_n,2}\phi}{\mathcal{A}_{U_n,2}}\right)\right)} \sum_{r=0}^{m_{U_n,E}+p-1} \binom{m_{U_n,E}+p-1}{r} \\ \times \int_0^\infty t^{m_{U_n,2}-r-1} e^{-\left(\left(\frac{m_{U_n,E}}{\mathcal{A}_{U_n,E}} \frac{t}{\phi A_2 - A_3}\right) + \left(\frac{m_{U_n,2}}{\mathcal{A}_{U_n,E}} \frac{\phi}{t\phi A_2 - A_3}\right)\right)} dt \Bigg]. \end{aligned} \quad (31)$$

Finally, we obtain the final expression of $\mathcal{P}_{\text{out}}$ by using [49, eq. (3.471.9)], which is expressed in (10).

Appendix B

First, we assume that $Z = |h_{B,1}|^2$ follows gamma distribution, whereas $V = |h_{1,U_n}|^2$ follows composite channel, i.e., log-normal and gamma distribution. By employing (4) and (12), the CDF $F_{\gamma_{1,U_n}}(\psi)$ can be expressed as

$$\begin{aligned} F_{\gamma_{1,U_n}}(\psi) &= \Pr\left[\frac{B_1 VZ}{B_2 VZ + B_3} < \psi\right] \\ &= \int_0^\infty F_Z\left(\frac{B_3 \psi}{(B_1 - B_2\psi)v}\right) f_V(v) dv \end{aligned} \quad (32)$$

where $B_1 = (\eta\alpha P_B/N_0)$, $B_2 = (\eta\alpha k^2 P_B/N_0)$, and $B_3 = \beta(1-\alpha)$. Then, substitute the CDF of the composite channel distribution and the PDF expressed in (2) into (32), and subsequently perform the necessary mathematical simplifications using [49, eqs. (3.381.3), (3.471.9)], we derive $F_{\gamma_{1,U_n}}(\psi)$ as expressed in (13).

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Gaurav Kumar Pandey (Student Member, IEEE) received the M.Tech. degree in communication engineering from the National Institute of Technology Nagaland, Dimapur, India, in 2018. He is currently pursuing the Ph.D. degree with the Department of Electronics and Communication Engineering, National Institute of Technology Silchar, Silchar, India.

His research interests include green communications, UAV communications, physical layer security, and optimization techniques.

Dr. Pandey is a recipient of a GATE Fellowship for pursuing a Ph.D. He received the Best Paper Awards from the IEEE SILCON 2023, IEEE ANTS 2023, and IEEE CICT 2024. He is serving as a Reviewer in various IEEE journals and conferences, including IEEE TRANSACTIONS ON GREEN COMMUNICATIONS AND NETWORKING and IEEE International Students' Conference on Electrical, Electronics and Computer Sciences.

Devendra Singh Gurjar (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from the Indian Institute of Technology Indore, Indore, India, in 2017.

He was with the Department of Electrical and Computer Engineering, University of Saskatchewan, Saskatoon, SK, Canada, as a Postdoctoral Research Fellow. He is currently an Assistant Professor with the Department of Electronics and Communication Engineering, National Institute of Technology Silchar, Silchar, India. With this fellowship, he wa with the Department of Information Security and Communication Technology, Norwegian University of Science and Technology, Trondheim, Norway. His research interests include MIMO communication systems, cooperative relaying, device-to-device communications, smart grid communications, physical layer security, and green communications.

Dr. Gurjar is the recipient of the Alain Bensoussan Fellowship 2019 from the European Research Consortium for Informatics and Mathematics. He is a member of the IEEE Communications Society and the IEEE Vehicular Technology Society.

Suneel Yadav (Senior Member, IEEE) received the Ph.D. degree in electrical engineering with the Indian Institute of Technology Indore, Indore, India, in 2016.

He is currently with the Department of Electronics and Communication Engineering, Indian Institute of Information and Technology Allahabad, Prayagraj, India, as an Assistant Professor, where he is serving as a Faculty In-Charge of Mobile and Wireless Networking Laboratory. He has numerous publications in peer-reviewed journals and conferences. His current research interests include wireless relaying techniques, cooperative communications, cognitive relaying networks, device-to-device communications, reconfigurable intelligent surfaces, signal processing, physical layer security, and MIMO systems.

Dr. Yadav also received the ICT Express Best Reviewer Award 2023 from the Korean Institute of Information and Communication Sciences. He is serving as a Reviewer in a number of international journals, including the IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY and the IEEE COMMUNICATIONS LETTERS.

Dragana Krstic received the B.Sc., M.Sc., and Ph.D. degrees from the Faculty of Electronic Engineering, University of Nis, Niš, Serbia, in 1990, 1998, and 2006, respectively.

She has been with the Faculty of Electronic Engineering, University of Nis, since 1990. Her fields of interest include wireless and mobile telecommunication systems. As the author/a co-author, she wrote over 320 scientific papers.

Dr. Krstic is an editor/a member of editorial board of a few international journals and reviewer in a number of international journals, including the IEEE TRANSACTIONS ON COMMUNICATIONS, IEEE COMMUNICATIONS LETTERS, IEEE ACCESS, IEEE WIRELESS COMMUNICATIONS LETTERS, IEEE SENSORS JOURNAL, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, and IEEE Open Journal of the Communications Society. She is also a member of the Technical Program Committee for 230 conferences and beside these a reviewer for over 220 others.

Yuming Jiang (Senior Member, IEEE) received the Ph.D. degree from the National University of Singapore (NUS), Singapore, in 2001.

Before his Ph.D. study, he was with Motorola, Beijing, China. He was a Research Engineer with NUS, from 1999 to 2001. From 2001 to 2003, he was with the Institute for Infocomm Research, Singapore, as a member of Technical Staff and a Research Scientist. In January 2004, he joined the Norwegian University of Science and Technology, Trondheim, Norway, where he has been a Full Professor since December 2005. He visited Northwestern University, Evanston, IL, USA, from August 2009 to July 2010, and also with Columbia University, New York, NY, USA, from August 2015 to July 2016. His research interests are in the provision and analysis of quality of service guarantees in communication networks, network management, and network calculus. In addition, a recent effort is on investigating basic properties of the traffic and service models adopted in time-sensitive networking and deterministic networking.

Prof. Jiang is recipient of a Fellowship from the European Research Consortium for Informatics and Mathematics. He was the Chair and the Board Member of IEEE Norway Section. He was the Co-Chair of IEEE Globecom2005—General Conference Symposium, the TPC Co-Chair of 67th IEEE Vehicular Technology Conference 2008, the General/TPC Co-Chair of International Symposium on Wireless Communication Systems from 2007 to 2010, the Steering Committee Chair of ISWCS since 2009, the General Chair of IFIP Networking 2014 Conference, the Chair of 2018 International Workshop on Network Calculus and Applications (NetCal 2018), and the TPC Co-Chair of the 32nd International Teletraffic Congress (ITC32), 2020.

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